



Dilshan Sanjaya

Instrumentation & Automation Engineer



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[Engineering Projects Portfolio](#)



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277/A, Hamlet 02, South Bank, Lunugamwehera

PROFESSIONAL SUMMARY

Motivated Instrumentation and Automation Technology degree candidate with hands-on experience in industrial automation, electrical, electronic and IoT systems. Skilled in PLC programming, sensor integration, PCB design, microcontroller-based development, and industrial monitoring systems. Through industrial training and academic projects, I designed and implemented automated packet tracking, temperature monitoring, and IoT based data acquisition systems that deliver reliable hardware software integration.

EDUCATION

- Faculty of Technology, University of Colombo** 2021 – Present
Bachelor of Engineering Technology (Hons) in Instrumentation and Automation Technology
Status: All coursework and industrial training completed; awaiting graduation.
- H/St. Mary's College** 2018 – 2020
GCE (A/L) Engineering Technology Stream: Science for Technology (A), Engineering Technology (B), ICT (C) | Z-score: 2.1275

EXPERIENCE

- Cargills Convenient Foods Ltd (Engineering Trainee)** Feb 2025 - July 2025
Supported factory maintenance activities, applied 5S principles, and designed mechanical components using SOLIDWORKS. Developed an ESP32-S3 packet counting and PID control system with a real-time Firebase and React web dashboard. Applied electrical wiring and structural design skills by building Vesak lanterns with custom lighting sequences.
- Greenery Tech (Pvt) Ltd (Trainee Technical Assistant)** Jan 2022 - July 2022
Assisted with the installation and maintenance of industrial Reverse Osmosis (RO) plant machinery to support continuous factory operation. Troubleshoot and repaired domestic water purification systems, providing hands-on technical support and improving client communication skills.

TECHNICAL SKILLS

- Industrial Automation:** PLC Programming (Siemens), Control System Design, Pneumatic System Design, Power Electronics, Motor Control & Protection, Safety Interlocks.
- Electrical & Electronic Systems:** Three-Phase Circuits, Panel Board Assembly, Relay Logic, Circuit Simulation (SimulIDE), PCB Design (EasyEDA Pro), Signal Conditioning.
- CAD & 3D Modeling:** Mechanical Component Design, AutoCAD, SOLIDWORKS.
- Embedded Systems & IoT:** Microcontrollers (ESP32, Arduino, ATmega328P, ATtiny85).
- Software Development:** React, React Native, Node.js, Firebase, Git, GitHub.
- Office Applications:** Microsoft Word, Excel, PowerPoint.

PROJECTS

- **Research Project - Plant Electrophysiological Signal Analysis under Different Salinity Level** November 2025 - June 2026
Designed data acquisition circuit to analyze electrophysiological signals of five tomato cultivars under salinity stress. Developed a feature-extraction machine learning model to identify stress levels and evaluate the most suitable cultivar.
- **PID Monitoring System for Cooling Room** March 2025 - July 2025
Built a real-time PID monitoring system for a cooling room using an ESP32-S3 board to track and control temperature. Created a web-based system to display live sensor data and monitor the cooling process efficiently.
- **Automated Sausage Packet Counting System** March 2025 - June 2025
Built an automated packet counting system using an ESP32-S3, a laser beam, and a barcode reader to accurately track production line output. Created a web-based display system integrated with Firebase to monitor daily counts in real-time and improve packaging efficiency.
- **Dual-Mode Star-Delta Starter (Manual/PLC)** October 2025 - December 2025
Built a three-phase motor starter panel as a group project, featuring both manual relay and automated PLC control modes. Programmed the ladder logic and wired critical safety features, including mechanical interlocks and an emergency stop, to protect the equipment.
- **Data Acquisition System** November 2023 - December 2023
Developed a real-time data acquisition (DAQ) system using an ESP32 to monitor smart greenhouse sensors.

SOFT SKILLS

- Problem-Solving
- Teamwork and Collaboration
- Time Management
- Self-Learning
- Effective Communication

LANGUAGES

- Sinhala
- English

REFERENCES

Dr. B. L. Sanjaya Thilakarathne

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